1. Combine Two Tables

```
SELECT FirstName, LastName, City, State
FROM Person
LEFT JOIN Address ON Person.PersonId = Address.PersonId;
```

2.Employees Earning More Than Their Managers

```
SELECT a.name AS Employee
FROM Employee a
LEFT JOIN Employee b ON a.ManagerId = b.Id
WHERE a.Salary > b.Salary;
```

3.Duplicate Emails

```
SELECT Email
FROM Person_1
GROUP BY Email
HAVING COUNT(*) > 1;
```

4.Customers Who Never Order

```
SELECT Name AS Customers
FROM Customers
WHERE Id NOT IN (SELECT CustomerId FROM Orders);
```

5.Delete Duplicate Emails

```
DELETE p1
FROM Person p1, Person p2
WHERE p1.Email = p2.Email AND p1.Id > p2.Id;
```

6.Invalid Tweets

```
SELECT tweet_id
FROM Tweets
WHERE LENGTH(content) > 15;
```

7.Recyclable and Low Fat Products

```
SELECT product_id
FROM Products
WHERE low_fats='Y' AND recyclable='Y';
```

8.Find Customer Referee

```
SELECT name
FROM customer
WHERE referee_id != 2 OR referee_id IS NULL;
```

9.Replace Employee ID With The Unique Identifier

```
SELECT unique_id, name
FROM Employees
LEFT JOIN EmployeeUNI ON Employees.id = EmployeeUNI.id;
```

10.Big Countries

```
SELECT name, population, area
FROM World
WHERE area >= 3000000 OR population >= 25000000;
```

11.Second Highest Salary

```
SELECT (
  SELECT DISTINCT Salary
  FROM Employee
  ORDER BY Salary DESC
  LIMIT 1 OFFSET 1
) AS SecondHighestSalary;
```

12.Product Sales Analysis II

```
SELECT product_id, SUM(quantity) AS total_quantity
FROM Sales
GROUP BY product_id;
```

13.Unique Orders and Customers Per Month

```
SELECT LEFT(order_date, 7) AS month,
COUNT(DISTINCT order_id) AS order_count,
COUNT(DISTINCT customer_id) AS customer_count
FROM Orders
WHERE invoice > 20
GROUP BY month;
```

14.Find Total Time Spent by Each Employee

```
SELECT event_day AS day, emp_id,
SUM(out_time - in_time) AS total_time
FROM Employees
GROUP BY day, emp_id;
```

15.User Activity for the Past 30 Days I

```
SELECT activity_date AS day, COUNT(DISTINCT user_id) AS active_users
FROM Activity
WHERE activity_date BETWEEN DATE_ADD('2019-07-27', INTERVAL -29 DAY)
AND '2019-07-27'
GROUP BY activity_date;
```

16.Classes More Than 5 Students

```
SELECT class
FROM courses
GROUP BY class
HAVING COUNT(DISTINCT student) >= 5;
```

17. Actors and Directors Who Cooperated At Least Three Times

```
SELECT actor_id, director_id
FROM ActorDirector
GROUP BY actor_id, director_id
HAVING COUNT(*) >= 3;
```

18.Group Sold Products By The Date

```
SELECT sell_date,
COUNT(DISTINCT product) AS num_sold,
GROUP_CONCAT(DISTINCT product) AS products
FROM Activities
GROUP BY sell_date
ORDER BY sell_date;
```

19.Students With Invalid Department

```
SELECT id, name
FROM Students
WHERE department_id NOT IN (SELECT id FROM Departments);
```